



Insulinemic energy and amino acids supplied postruminally independently regulate the performance and metabolism of peak-lactation dairy cows

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ABSTRACT

Increasing AA sequestration in milk is essential for improving N efficiency in dairy cows and to reduce the environmental impacts of dairy production. Isoleucine, Leu, and Met are promising AA to improve milk protein yield and N efficiency due to their signaling roles in mammary epithelial cells. Glucogenic diets can also stimulate milk protein synthesis and, through their effect on insulin, potentiate the signaling effects of AA. Our objective was to evaluate the effects of Ile, Leu, and Met on production performance and N metabolism, and the role of energy source (ES; glucose or acetate) on lactating cows' responses to AA. The experiment consisted of a 4 × 4 Latin square design replicated 4 times with 4 periods of 21 d and 4 treatments arranged as a 2 × 2 factorial. Sixteen ruminally cannulated, multiparous, Holstein cows were housed in tiestalls and fed the same experimental diet with 4% energy and 15% MP deficiency. The 4 treatments were continuously infused into the abomasum and consisted of isoenergetic amounts (2.86 Mcal/d) of glucose (GLC) or acetate (ACE), without AA as negative control, or with a mix of 12 g/d of Ile, 50 g/d of Leu, and 20 g/d of Met (ILM). Experimental periods consisted of 2 wk of adaptation and 1 wk of sampling, in which feed, urine, milk, and blood samples were collected. Data were analyzed with a mixed model in R, containing the fixed effects of period, square, AA infusion, ES, and ES × AA interaction, and the random effects of cow within square. Overall, ES did not alter the response to AA on production or N metabolism (i.e., no significant interactions). Infusion of ILM, regardless of ES, increased circulating insulin concentration, milk protein and fat contents, and yields of de novo and mixed fatty acids, as predicted by Fourier-transform mid-infrared spectroscopy, but

decreased lactose yield and concentration and tended to decrease milk yield. Meanwhile, ILM did not alter ECM or N efficiency for milk production. Compared with ACE, GLC increased DMI, circulating concentrations of glucose and insulin in plasma, and yields of milk and milk protein, but decreased circulating levels of acetate, BHB, and nonesterified fatty acids, as well as yields of presynthesized fatty acids, total milk fat, and fat-corrected milk. Similar to AA infusion, ES did not affect ECM yield nor N efficiency for milk production. In summary, compared with ACE, GLC did not alter the performance response to ILM but independently affected production and metabolism, consistent with its insulinemic potential. Cows responded mildly to AA infusion, most remarkably increasing de novo milk fatty acid production and milk protein content without impairing N efficiency, while also decreasing milk lactose concentration and yield.

Key words: amino acid, glucose, acetate, insulin, N efficiency

INTRODUCTION

Ruminant animals play an important role in agricultural sustainability due to their ability to convert human-inedible or low-quality ingredients into highly nutritious products such as milk and meat. However, the efficiency of dairy cattle to convert feed N into milk N is relatively low, ranging from 16% to 36% (Powell et al., 2010), leading to its excretion in manure, which raises environmental concerns of eutrophication, greenhouse gas pollution (nitrous oxide), and particulate matter formation (Hristov et al., 2011). Furthermore, the modern, high-producing dairy cow requires a large amount of RUP, which is the most expensive protein source (French, 2024), to complement the ruminal microbial protein and meet its metabolizable AA requirements. Dijkstra et al. (2013) suggested that the theoretical upper limit of N use efficiency (NUE) in dairy cows is 43%, and the reported upper quartiles of NUE for the United States and European Union are 33% and 32%, respectively (Calsamiglia

Received October 14, 2025.

Accepted January 8, 2026.

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

et al., 2010; Dijkstra et al., 2013), indicating that further improvements are possible.

The roles of individual AA to stimulate milk protein synthesis and improve NUE have been explored extensively, although with many challenges remaining. The most recent edition of *Nutrient Requirements for Dairy Cattle* (NASEM, 2021) considers independent effects of His, Ile, Leu, Lys, and Met on milk protein yield, challenging the traditional concept that a single limiting AA would hinder the effects of any other AA. The regulatory mechanisms of those 5 AA have not been fully elucidated. However, Ile, Leu, and Met have shown stimulatory effects on the mechanistic target of rapamycin (**mTOR**) signaling pathway in cultured mammary epithelial cells (Appuhamy et al., 2012; Arriola Apelo et al., 2014b; Pszczolkowski et al., 2020). The mTOR protein complex 1 (**mTORC1**) is a central hub of mammary cell metabolism and a major regulator of milk protein translation (Moshel et al., 2006; Pszczolkowski and Arriola Apelo, 2020). Furthermore, Pszczolkowski et al. (2020) showed that the stimulation of mTORC1 by Ile, Leu, and Met is potentiated by insulin in cultured mammary epithelial cells. In dairy cows, insulin (Mackle et al., 2000) and insulinemic energy sources such as glucose (Vanhatalo et al., 2003; Danes et al., 2020; Omphalius et al., 2020) and propionate (Raggio et al., 2006; Lemosquet et al., 2009) are able to stimulate milk protein production even when AA are deficient. However, the relationship between insulin or insulinemic energy and AA or protein on milk protein synthesis regulation is inconclusive. In the study by Mackle et al. (2000), casein plus branched-chain AA (**BCAA**) had no effect on milk protein production except under a hyperinsulinemic-euglycemic clamp. Rius et al. (2010) observed a negative effect of abomasal casein infusion alone, but combined with starch it increased milk protein yield compared with the water control or starch or casein alone. Conversely, Danes et al. (2020), who had seen an effect of glucose on milk protein production, failed to see further improvement when casein was infused alone. Finally, Curtis et al. (2018), who observed an effect of intravenous glucose infusions, reported a negative milk protein response to the addition of the 3 BCAA. Most studies evaluating energy and protein interactions have used casein, or as in the case of Curtis et al. (2018), BCAA. Excess of nonlimiting AA (Nichols et al., 2019b) or imbalances among EAA (Bequette et al., 2000) can reduce mammary clearance of AA, giving room for other factors such as type and amount of nutrient infused, background nutritional status, or physiological state to alter the relationship between insulinemic energy and AA. Furthermore, site of infusion could alter incretin hormones production and insulin secretion (Gameiro et al., 2005; Baumgard et al., 2016). By stimulating the protein synthesis machinery at mammary level, mTORC1-stimulating AA could increase mammary demand for AA

Table 1. Ingredients and observed nutrient composition of the experimental diet

Item	DM, %
Ingredient	
Alfalfa haylage	17.7
Corn silage	35.3
Dry corn grain	15.5
Soybean hulls	17.2
Soybean meal, expeller ¹	6.9
Fatty acid blend ²	1.5
Urea	0.87
Calcium carbonate	2.06
Magnesium oxide	1.38
Mineral premix ³	0.41
Sodium bicarbonate	0.86
Rumen-protected lysine ⁴	0.34
Nutritional composition	
CP	16.0
RDP ⁵	11.1
RUP ⁵	4.9
MP ⁵	8.91
Forage NDF	18.7
NDF	33.5
ADF	22.9
Starch	26.3
Fat	4.31
ME, ⁵ Mcal/kg of DM	2.56

¹SoyPlus (Landus Cooperative, Ames, IA).

²Spectrum Fusion (Perdue AgriBusiness, Salisbury, MD).

³Provided per kg of DM: 117 mg of Zn, 81 mg of Mn, 8 mg of Fe, 28 mg of Cu, 2 mg of I, 0.4 mg of Co, 0.4 mg of Se, 10.83 kIU of vitamin A, 2.17 kIU of vitamin D, and 54.16 IU of vitamin E.

⁴AjiPro-L (Ajinomoto Health and Nutrition North America, Itasca, IL).

⁵Estimated using NASEM (2021).

(Pszczolkowski and Arriola Apelo, 2020), whereas insulin would be required to maximize this effect (Pszczolkowski et al., 2020). Therefore, we hypothesize that abomasal infusions of glucose, as an insulinemic energy source, would potentiate the effects of Ile, Leu, and Met on milk protein production and improve NUE compared with acetate.

MATERIALS AND METHODS

Animals and Experimental Design

The experiment was carried out at the University of Wisconsin–Madison Dairy Cattle Center (Madison, WI). Sixteen ruminally cannulated, multiparous Holstein cows were selected for the study. Before the trial, cows were acclimated to the tiestall facility for at least 2 wk. As standard facility management, cows were milked twice daily, at 0400 h and 1600 h, fed once after the morning milking, and had constant access to fresh, clean water. At the beginning of the trial, cows averaged ($\pm$ SD) 70.4 $\pm$ 26.5 DIM, 50.6 $\pm$ 4.5 kg/d milk yield, and 647.9 $\pm$ 45.0 kg BW.

The experiment consisted of a 4 $\times$ 4 Latin square replicated 4 times, with 4 periods and 4 treatments balanced for carryover effects and arranged as a 2 $\times$ 2 factorial.

Table 2. Individual ingredients infused daily by treatment¹

Ingredient, g	GLC-CTL	GLC-ILM	ACE-CTL	ACE-ILM
Glucose	766	766	0	0
Glacial acetic acid	0	0	391	391
Sodium acetate	0	0	561	561
Sodium chloride	400	400	0	0
Ile	0	12	0	12
Leu	0	50	0	50
Met	0	20	0	20

¹All treatments were dissolved in 8 L of tap water no more than 24 h before infusing.

The 4 squares were divided into 2 groups that were run consecutively, with squares within groups run concurrently. Cows were grouped into squares according to DIM and randomly assigned to treatments within square. Experimental periods were 21 d long, consisting of 2 wk of adaptation to the treatment, and 1 wk of sampling. A single TMR was formulated for all cows using Dairy-8 software (NASEM, 2021) to meet 85% of MP requirement and 96% of ME requirement, and was fed targeting 10% refusals. The characteristics of the lactating Holstein cow used to formulate the diet included 750 kg BW, 28.5 kg/d DMI, 44 kg/d milk yield, 3.8% milk fat, 3.0% milk true protein, and 4.6% milk lactose. The ingredients and nutrient composition of the diet are shown in Table 1. During the third week of each period, the feed was divided into 6 equal meals offered every 3 h (except for afternoon milking time) until midnight, aiming for a more homogeneous nutrient supply along the day.

The 2 treatment factors were energy source (ES) and AA infusion. The 2 ES were glucose (GLC, 4.25 mol/d) or acetate (ACE, 13.7 mol/d), provided as an isoenergetic control (2.86 Mcal/d), considering their respective energy densities of 673 and 209 kcal/mol (Danes et al., 2020). The ACE solution was buffered to a pH of 4.7 by mixing equimolar amounts of sodium acetate and glacial acetic acid (99.9%, Lab Alley, Austin, TX), resulting in a total of 13.7 mol of ACE infused per day. Sodium chloride was added to the GLC infusion treatments to provide equimolar amounts of Na between ES. The 2 AA factor levels were a control with no AA infused (CTL) or a mix of 12 g/d of Ile, 50 g/d of Leu, and 20 g/d of Met (ILM). The amounts of Ile, Leu, and Met infused in the ILM treatment were based on NASEM (2021) predictions of metabolizable contents of those AA in the diet, the level of production prior to trial (44 kg/d of milk), and NASEM (2021) target metabolizable efficiency for each of those AA, plus a 7.5% excess to buffer potential differences in DMI between treatments. Treatment solutions were prepared daily by mixing AA and ES in 8 L of tap water (Table 2), with refrigeration until use. Treatment solutions were continuously infused into the abomasum, except when cows were taken to the parlor for milking (total of ~20 h/d infusion), and replaced daily after the

morning milking. A 16-channel peristaltic pump (Watson Marlow Fluid Technology Solutions, Marlow, UK), connected to individual infusion lines was used to deliver the treatment solutions into the abomasum. The integrity and correct placement of the abomasal infusion lines were checked daily, before infusions started. Feed intake, milk production, and infusion start times were recorded daily throughout the study. Body weight was recorded weekly after morning milking.

Sample Collection

Samples of TMR, concentrate, and forages offered on d 16 to 18 were compiled and stored at -20°C until further analysis. Milk samples were collected at every milking from d 15 p.m. until d 19 a.m. and refrigerated with 2-bromo-2-nitropropane-1,3-diol as preservative until analysis of milk component by mid-infrared spectroscopy (Ag-Source Laboratory, Marshfield, WI). Blood samples were collected from a coccygeal vessel into heparinized tubes, for plasma isolation, and serum-separating tubes, from d 17 to 19, every 3 h between a.m. and p.m. milkings, staggered to represent 10 hourly samples between 0700 h and 1600 h. Blood tubes were refrigerated (heparin) and plasma or serum isolated by centrifugation (2,500 × g, 4°C, 15 min) within 2 h of collection. Aliquots of plasma and serum were stored at -80°C until further analysis. Spot urine samples were collected on d 17 to 19 every 6 h, staggered to represent a total of 12 samples collected every 2 h in a 24-h period. Urination was stimulated by massaging the lower perivulvar region. A 4-mL urine aliquot was immediately acidified with 36 mL of 0.144 N H₂SO₄. The 12 acidified samples were further aliquoted, pooled, and stored at -20°C until further analysis.

Laboratory Analyses

Feed samples were compiled per period and dried at 60°C for 72 h to determine DM concentration. These samples were then ground to pass a 1-mm screen (Wiley Mill, A. H. Thomas Company, Philadelphia, PA) and analyzed in a commercial laboratory (DairyLand Laboratories Inc., Arcadia, WI) for N by combustion

(official method 990.03; AOAC, 2012); total fat was analyzed by acid hydrolysis followed by ether extraction using a SoxCap 2047 in combination with Soxtec extraction systems (Foss Analytics, Eden Prairie, MN); amylase-treated NDF and ADF content were analyzed based on AOAC official methods 2002.04 and 973.18, respectively (AOAC, 1997, 2006); and starch using the acetate buffer method described by Hall (2015), modified to measure glucose with a YSI Biochemistry Analyzer (Xylem Inc., Yellow Springs, OH). All plasma samples were individually analyzed for insulin (Bovine Insulin ELISA Kit, Mercodia, Uppsala, Sweden), glucose (Mutarotase-GOD method, Wako Autokit Glucose, Fujifilm, Lexington, MA), acetate, and AA. Plasma acetate concentration was determined by GC-MS following the protocol outlined by Rohde et al. (2022), using ^4D -acetate as internal standard. Plasma AA were analyzed by liquid chromatography–MS as previously described (Hoppmann and Arriola Apelo, 2024) with universally labeled ^{13}C AA as internal standards. Serum nonesterified fatty acids (NEFA), BHB, triglycerides (TG), and urea N (UN) were analyzed with a chemical analyzer (Catachem, Oxford, CT) using a NEFA-HR(2) kit (Wako, Fujifilm, Lexington, MA), reagents from Catachem for BHB, from MedTest Dx (Pointe Brand, Canton, MI) for TG, and from Arbor Assays (Ann Arbor, MI) for UN. Urinary UN and urine creatinine were determined using a Lachat QuikChem8500 flow injection analysis system (Hach Company, Lachat Instruments Division, Loveland, Colorado).

Calculations

Average daily gain was calculated as the slope of the regression of weekly BW measurements per cow, per period (Liu et al., 2021). Energy-corrected milk was calculated as $[(0.327 \times \text{kg of milk}) + (12.95 \times \text{kg of milk fat}) + (7.20 \times \text{kg of milk protein})]$, and 3.5% FCM was calculated as $[(0.4324 \times \text{kg of milk}) + (16.216 \times \text{kg of milk fat})]$, as denoted by DRMS (Milk 2014). Due to issues with N analysis in urine, total urinary N (g/d) was calculated as $\text{MUN (mg/dL)} \times \text{BW (kg)} \times 0.026$, according to Kohn et al. (2002), and NUE was calculated as $\text{milk true protein} \div 6.38$ (TP; g/d) $\div$ [N intake + N infused (g/d)].

We used an R version of Dairy-8 (NASEM, 2021) to estimate N partitioning in individual cows, considering their observed DMI, BW, and productive performance. Using NASEM (2021) predictions, ruminal N losses were calculated as RDP balance (g)/6.25; intestinal or digestive N losses as [RUP + duodenal microbial CP (g) – MP not including infusions (g)]/6.25; and catabolic N losses as [MP-N including infusions (g) – scurf TP-N (g) – fecal endogenous TP-N (g) – milk TP-N (g)]. Efficiency of individual AA was determined ac-

cording to Lapierre et al. (2020), on the same R version of Dairy 8 (NASEM, 2021).

Statistical Analysis

Data were analyzed using the following linear mixed model:

$$Y_{ijkl} = \mu + S_h + C_{i(h)} + P_{j(h)} + E_k + A_l + EA_{kl} + \varepsilon_{ijkl},$$

where Y_{ijkl} = dependent variable, μ = overall mean, S_h = fixed effect of square, $C_{i(h)}$ = random effect of cow within square, $P_{j(h)}$ = fixed effect of period within square, E_k = fixed effect of ES, A_l = fixed effect of AA infusion, EA_{kl} = interaction between ES $\times$ AA, and ε_{ijkl} = residual error. Data were analyzed in R Studio v. 2023.06.1 (Posit Software, PBC), using the lme4 package to fit the model. Where applicable, variables with multiple time points were averaged per cow, per period, before statistical analysis ($n = 64$). Following veterinarian recommendations, one cow had to be discontinued from the infusions in her second period and another cow in her last period due to extended reductions in DMI and milk production that compromised their health and wellbeing. The former cow was replaced with another to complete the remaining 2 periods. The 3 missing experimental units represented treatments GLC-CTL, ACE-CTL, and ACE-CTL, and their values were estimated and imputed according to Kramer and Glass (1960), to ensure the Latin squares were balanced. Significance was declared at $P \leq 0.05$ and trends at $0.05 < P \leq 0.1$.

RESULTS

Animal Performance and Blood Metabolites

The supplies of NEL, MP, and metabolizable (M-) AA for each treatment as predicted by NASEM (2021) using observed intake and production data are shown in Table 3. Numerically, the supplies of NEL (46.2 Mcal/d), MP (2,320 g/d), M-His (55.5 g/d), M-Ile (141 g/d), M-Leu (209 g/d), M-Lys (196 g/d), and M-Met (51 g/d) were greater for GLC compared with ACE (44.4 Mcal/d, 2,249 g/d, 54 g/d, 137 g/d, 203 g/d, 190 g/d, and 49 g/d, respectively).

The effects of treatments on DMI, ADG, milk production, and milk composition are shown in Table 4. No interaction was observed on any of these variables. Relative to ACE, GLC increased DMI by 1.1 kg ($P = 0.01$), and milk yield by 2 kg/d ($P < 0.001$), but 3.5% FCM decreased by 1.0 kg/d. No effects of ES or AA were observed on ECM ($P > 0.05$). Average daily gain tended ($P = 0.06$) to be greater for GLC (250 g/d) compared with ACE (40 g/d). Milk lactose content was decreased by ILM infusion ($P = 0.003$), and lactose yield was independently decreased by

Table 3. Supplies of NEL, MP, and metabolizable AA (M-AA), as predicted by NASEM (2021)^{1,2}

Item	Treatment ³			
	GLC		ACE	
	CTL	ILM	CTL	ILM
NEL, Mcal/d				
Supply	45.9	46.4	44.3	44.5
Balance	3.4	3.6	4.7	3.8
MP, g/d				
Diet supply	2,308	2,332	2,252	2,246
Infusion supply	0	82	0	82
M-AA, g/d				
His				
Diet supply	55	56	54	54
Ile				
Diet supply	140	142	137	137
Infusion supply	0	12	0	12
Leu				
Diet supply	208	210	203	202
Infusion supply	0	50	0	50
Lys				
Diet supply	195	197	190	190
Met				
Diet supply	51	51	49	49
Infusion supply	0	20	0	20

¹Supplies and balance of NEL estimated by NASEM (2021) using observed DMI, BW, milk yield, and milk components of individual cows.

²NEL balance = NEL supply (Mcal/d) – NEL for maintenance and lactation.

³GLC = infusions including 4.25 mol/d of glucose; ACE = infusions including 13.7 mol/d of acetate; CTL = infusions not including AA; ILM = infusions including 12, 50, and 20 g/d of Ile, Leu, and Met, respectively.

infusion of ILM ($P = 0.03$) and ACE compared with GLC ($P < 0.001$). Milk protein content was greater for ILM ($P < 0.001$) and GLC-infused cows ($P = 0.044$) relative to CTL and ACE, respectively. However, the effect of ILM on protein content was not translated into an increase in protein yield ($P = 0.32$) due to a tendency ($P = 0.09$) of ILM to decrease milk yield. Meanwhile, GLC increased milk protein yield (MPY) by 72 g/d relative to ACE ($P < 0.001$). Milk fat content was increased 0.48 percentage points by ACE ($P < 0.001$) and 0.14 percentage points by ILM ($P = 0.04$), but only ACE infusion increased fat yield (+0.114 kg/d) and FCM yield (+1 kg/d, $P = 0.008$). However, when considering the yield of fatty acids (FA) by length, ILM treatment increased the yield of FA shorter than 16 C (+14 g/d, $P = 0.013$) and tended to increase the yield of FA equal to 16 C (+14 g/d, $P = 0.076$), compared with CTL. Compared with GLC, ACE treatment increased the yield of 16 C FA (+19 g/d, $P = 0.018$) and FA longer than 16 C (+101 g/d, $P < 0.001$).

Circulating concentrations of metabolites in plasma and serum are shown in Table 5. Infusions of GLC successfully increased ($P < 0.001$) the concentration of glucose in plasma (3.97 mM) relative to ACE (3.83 mM), although to a greater extent with ILM than with CTL, as indicated by the ES $\times$ AA interaction ($P = 0.05$). Furthermore, ACE infusion increased ($P <$

0.001) acetate plasma concentration (1.51 mM) relative to GLC (1.03 mM). Insulin plasma concentration was greater for GLC (0.58 μ g/L) than for ACE-infused cows (0.45 μ g/L; $P < 0.001$). In line, serum NEFA (0.12 mM) and BHB (11.3 mg/dL) concentrations were higher for ACE than for GLC-infused cows (0.08 mM and 8.1 mg/dL, respectively; $P < 0.001$). Additionally, ILM independently increased circulating concentrations of insulin (0.55 μ g/L; $P = 0.004$) and acetate (1.35 mM; $P = 0.002$), and tended to increase BHB (10.1 mg/dL; $P = 0.10$) relative to CTL (0.48 μ g/L, 1.18 mM, and 9.2 mg/dL, respectively). Serum TG concentrations were not affected by either factor.

Compared with CTL, ILM treatment increased plasma Met (+30 μ M; $P < 0.001$) and Leu concentrations ($P < 0.001$), although Leu increased more for ACE-ILM (+52.4 μ M) than for GLC-ILM (+27.4 μ M) relative to CTL (ES $\times$ AA; $P = 0.009$). Isoleucine was not affected by AA but it was by ES (123.2 μ M for GLC vs. 155.4 μ M for ACE; $P < 0.001$). Beyond infused AA, plasma concentrations of total AA (2,085 μ M; $P = 0.008$), non-infused essential AA (EAA-ILM; 532 μ M; $P < 0.001$), and NEAA (1,236 μ M; $P < 0.001$) were lower for ILM compared with CTL (2,156, 569, and 1,334 μ M, respectively). For individual EAA, ILM decreased circulating plasma concentrations of His (18 μ M; $P < 0.001$),

Table 4. Effects of energy source (ES) and AA supply on DMI, ADG, and milk production in lactating Holstein cows¹

Item ²	Treatment				SEM	P-value		
	GLC		ACE			ES	AA	Int.
	CTL	ILM	CTL	ILM				
DMI, kg/d	26.1	26.3	25.1	25.1	0.51	0.01	0.80	0.83
ADG, g/d	249	251	−24.2	104	121.4	0.06	0.56	0.57
Yield, kg/d								
Milk	42.7	41.7	40.6	39.8	0.95	<0.001	0.09	0.89
ECM	41.7	41.9	42.2	42.1	0.69	0.40	0.83	0.66
3.5% FCM	42.9	43.0	44.1	43.8	0.70	0.008	0.86	0.49
Protein	1.146	1.155	1.068	1.089	0.026	<0.001	0.32	0.71
Lactose	2.049	1.988	1.946	1.899	0.050	<0.001	0.03	0.77
Fat	1.506	1.542	1.636	1.639	0.024	<0.001	0.24	0.31
FA <C ₁₆ , g/d	342	360	342	350	7.4	0.36	0.01	0.33
FA C ₁₆ , g/d	511	534	539	544	10.8	0.02	0.08	0.28
FA >C ₁₆ , g/d	543	544	641	649	12.7	<0.001	0.66	0.74
Composition, %								
Protein	2.69	2.77	2.64	2.75	0.04	0.04	<0.001	0.52
Lactose	4.81	4.77	4.79	4.77	0.04	0.80	0.003	0.37
Fat	3.54	3.71	4.05	4.16	0.08	<0.001	0.04	0.71
FA <C ₁₆	0.80	0.87	0.84	0.89	0.02	0.04	<0.001	0.55
FA C ₁₆	1.20	1.29	1.33	1.39	0.03	<0.001	0.02	0.64
FA >C ₁₆	1.27	1.31	1.59	1.65	0.05	<0.001	0.19	0.66

¹GLC = infusions including 4.25 mol/d of glucose; ACE = infusions including 13.7 mol/d of acetate; CTL = infusions not including AA; ILM = infusions including 12, 50, and 20 g/d of Ile, Leu, and Met, respectively; ES = effect of energy source; AA = effect of AA infusion; Int. = ES × AA interaction.

²FA <C₁₆, FA C₁₆, and FA >C₁₆ = milk fatty acids with less than, equal to, or more than 16 carbons in length, respectively.

Thr (96 μ M; $P = 0.014$), Trp (52 μ M; $P = 0.005$), and Val (166 μ M; $P < 0.001$) compared with CTL (24, 106, 55, 184 μ M, respectively). Meanwhile, compared with ACE, infusion of GLC decreased Arg (74 vs. 79 μ M; $P = 0.05$) and tended to decrease circulating plasma concentration of EAA-ILM (543 vs. 558 μ M; $P = 0.09$), but increased it for NEAA (1,334 vs. 1,236 μ M; $P < 0.001$). The responses of individual AA to the different ES were more erratic. Although Lys ($P = 0.009$) and Val ($P < 0.001$) decreased in response to GLC infusion (80 vs. 85 μ M, and 164 vs. 186 μ M for GLC vs. ACE, respectively), Thr increased (108 vs. 94.0 μ M). Among the NEAA, Asn (49 vs. 44 μ M; $P < 0.001$), Asp (3.2 vs. 3.0 μ M; $P = 0.004$), Gln (291 vs. 241 μ M; $P < 0.001$), Gly (405 vs. 379 μ M; $P = 0.004$), Ser (113 vs. 105 μ M; $P = 0.005$), and Tyr (82 vs. 75 μ M; $P = 0.003$) increased in response to GLC versus ACE, whereas Glu decreased (45 vs. 47 μ M; $P = 0.006$).

Nitrogen Metabolism and Efficiency

The effects of ES and AA on N metabolism, efficiencies, and estimated N losses are shown in Tables 6 and 7. Dietary N intake was greater for GLC ($P = 0.01$) due to its higher DMI, which also resulted in a numerically higher MP-N compared with ACE. Expectedly, ILM also increased MP-N relative to CTL, but the additive effects of ES and AA resulted in GLC-CTL having

MP-N similar to that of ACE-ILM. No effects of treatment were observed on urine N, MUN, or urine UN, but serum UN was decreased by ILM ($P < 0.001$) as well as urinary creatinine ($P = 0.04$). The efficiency of N use for milk protein production (NUE) was unaffected by either factor ($P > 0.13$). According to NASEM (2021) predictions, the greatest proportion of N losses were digestive (23.7% of total N supply), followed by catabolic losses (22%) and rumen losses (20.3%). Across treatments, rumen and digestive losses differed by less than 1 percentage points, but catabolic losses were 1.4 percentage points higher for ILM compared with CTL. Although not analyzed statistically, it is notable that the efficiencies of MP and all EAA were numerically higher for GLC compared with ACE, whereas slightly lower MP efficiency was observed for ILM compared with CTL. However, this was not the case for the efficiency of individual EAA except for Ile, Leu, and Met, for which ILM reduced their respective efficiencies by 0.05, 0.14, and 0.23 percentage units.

DISCUSSION

Typical lactation diets for dairy cows in North America, and particularly in the Midwest, consist mostly of corn and soybean feedstuffs, which are deficient in Lys and Met, respectively, relative to dairy cow requirements. Meanwhile, low-CP diets, and other diets that rely more

Table 5. Effects of energy source (ES) and AA supply on circulating concentrations of metabolites and AA in lactating Holstein cows¹

Item ²	Treatment				SEM	P-value		
	GLC		ACE			ES	AA	Int.
	CTL	ILM	CTL	ILM				
Insulin, µg/L	0.53	0.62	0.43	0.48	0.04	<0.001	<0.01	0.32
Glucose, mM	3.94	4.00	3.86	3.81	0.04	<0.001	0.90	0.05
Acetate, mM	0.98	1.07	1.38	1.63	0.06	<0.001	0.002	0.12
BHB, mg/dL	8.0	8.3	10.5	12.0	0.7	<0.001	0.10	0.28
NEFA, mM	0.08	0.09	0.12	0.13	0.01	<0.001	0.30	0.78
TG, mg/dL	10.0	9.8	10.1	10.3	0.4	0.20	0.95	0.24
AA, µM								
EAA								
Arg	75.1	73.8	77.8	79.3	3.3	0.05	0.98	0.51
His	24.2	18.2	23.9	18.1	1.8	0.84	<0.001	0.89
Ile	128	118	156	155	7	<0.001	0.31	0.41
Leu	84.8	112	102	155	5	<0.001	<0.001	0.009
Lys	81.6	78.4	84.4	84.8	3.0	<0.01	0.42	0.28
Met	16.8	47.4	17.8	46.1	1.8	0.95	<0.001	0.53
Phe	42.4	42.7	43.0	41.5	0.9	0.71	0.40	0.25
Thr	114	102	97.2	90.8	5.4	<0.001	0.014	0.42
Trp	54.4	51.9	54.6	51.3	1.5	0.86	0.005	0.65
Val	173	155	194	177	7	<0.001	<0.001	0.83
EAA-ILM ³	565	522	575	543	14	0.093	<0.001	0.52
NEAA								
Ala	273	262	275	255	12	0.79	0.09	0.61
Asn	50.4	47.3	45.2	43.1	1.4	<0.001	0.01	0.60
Asp	3.4	3.0	3.1	2.9	0.1	0.004	<0.001	0.03
Gln	312	276	251	231	8	<0.001	<0.001	0.21
Glu	45.8	43.5	47.3	47.6	1.4	0.006	0.29	0.19
Gly	423	386	389	370	19	0.004	0.002	0.29
Pro	82.9	77.5	79.9	74.4	3.0	0.12	0.007	0.98
Ser	116	110	110	98.7	5.0	0.005	0.001	0.35
Tyr	84.9	79.1	79.7	69.6	3.4	0.003	0.001	0.34
NEAA	1,392	1,284	1,280	1,192	32	<0.001	<0.001	0.65
Total AA	2,186	2,084	2,131	2,091	37	0.34	0.008	0.23

¹GLC = infusions including 4.25 mol/d of glucose; ACE = infusions including 13.7 mol/d of acetate; CTL = infusions not including AA; ILM = infusions including 12, 50, and 20 g/d of Ile, Leu, and Met, respectively; ES = effect of energy source; AA = effect of AA infusion; Int. = ES × AA interaction.

²TG = triglycerides; NEFA = nonesterified fatty acids. All metabolites were measured in plasma except BHB, NEFA, and TG, which were measured in serum.

³EAA-ILM does not consider infused AA. Calculated as the sum of Arg, His, Lys, Phe, Thr, Trp, and Val.

on microbial protein for their MP, tend to be deficient in His (Kim et al., 1999; Lee et al., 2012). Severe deficiencies of AA can compromise milk protein production due to the lack of substrate (Schwab et al., 1976; Bequette et al., 2000; Weekes et al., 2006). However, the recent update to the *Nutrient Requirements of Dairy Cattle* (NASEM, 2021) identifies Ile and Leu, along with Met, Lys, and His, as independent drivers of milk protein production (i.e., regardless of the level of other AA), suggesting substrate-independent effects of these 5 AA on regulatory mechanisms of milk protein synthesis (Arriola Apelo et al., 2014a). Otherwise, an AA would regulate milk protein production only when it is the most (first) limiting. Of the 5 AA included in NASEM's milk protein equation (Equation 6-6; NASEM, 2021), we selected the 3 with a common regulatory mechanism of protein synthesis regulation through the mTORC1 pathway, and,

given the effect of insulin on mTORC1, we explored the role of insulinemic energy on the effect of those AA in milk protein production. This rationale aligns with our hypothesis that insulinemic energy might enhance the anabolic signaling of these AA in vivo. All the cows were offered a single lactation diet, and despite an effect of ES on DMI, the increase in circulating concentrations of glucose and insulin in GLC, as well as increased concentration of acetate by ACE, confirmed an effective implementation of our infusion model. Furthermore, ILM did not affect DMI but did affect the circulating concentrations of Leu and Met. The concentration of Ile in plasma was not affected by ILM, possibly because it was infused at the smallest amount, because it was partially metabolized by the gastrointestinal tract (El-Kadi et al., 2006), because it was utilized by tissues more efficiently, or due to a combination of these.

Table 6. Effects of energy source (ES) and AA supply on N intake, metabolism, and use efficiency in lactating Holstein cows¹

Item	Treatment					<i>P</i> -value		
	GLC		ACE		SEM			
	CTL	ILM	CTL	ILM		ES	AA	Int.
Dietary N intake, g/d	671	676	646	647	13	0.01	0.80	0.83
Infused N, g/d	0.0	8.5	0.0	8.5	—	—	—	—
MP-N, g/d	371	386	359	372	—	—	—	—
Milk TP-N g/d	180	181	167	171	4	<0.001	0.32	0.71
Urine N, ² g/d	194	192	198	193	6.2	0.58	0.45	0.68
Urea N (UN)								
Serum UN, mg/dL	17.0	16.7	17.0	16.8	0.2	0.76	<0.01	0.44
Milk UN, mg/dL	11.5	11.5	11.7	11.4	0.3	0.80	0.53	0.61
Urine UN, g/d	124	124	126	124	3.8	0.68	0.62	0.71
Creatinine, mg/dL	60.4	59.4	62.0	57.3	1.6	0.83	0.04	0.16
NUE, ³ %	26.9	26.7	26	26.1	0.6	0.13	0.89	0.76

¹GLC = infusions including 4.25 mol/d of glucose; ACE = infusions including 13.7 mol/d of acetate; CTL = infusions not including AA; ILM = infusions including 12, 50, and 20 g/d of Ile, Leu, and Met, respectively; ES = effect of energy source; AA = effect of AA infusion; Int. = ES × AA interaction; TP = true protein.

²Total urine N (g/d) = MUN (mg/dL) × 0.026 × BW (kg), according to Kohn et al. (2002).

³N use efficiency = (milk true protein, g/d ÷ 6.38)/(N intake + N infused, g/d).

Energy Source Did Not Alter Metabolic and Production Responses to AA

Despite establishing a successful abomasal infusion model, and contrary to our hypothesis, ES did not affect the milk protein response to ILM compared with CTL. Likewise, no interaction effects were observed on DM and N intake, milk component yield and content, N partitioning, or circulating concentrations of most metabolites. The only interactions observed in this study were on circulating concentrations of glucose, Leu, and Asp, which responded to ILM infusion differently between GLC and ACE. Energy source had greater independent effects on production and metabolism than ILM, which had limited or even negative effects on production. This lack of production responses to ILM potentially masked any synergistic effects of GLC. Mackle et al. (2000) reported no effect of abomasal infusion of casein plus BCAA on MPY. Meanwhile, MPY increased by 25% (+213 g/d) when the casein treatment was combined with a hyperinsulinemic-euglycemic clamp (Mackle et al., 2000). Similar synergistic responses between insulin or insulinemic energy and protein were observed in another insulin clamp study combined with abomasal infusion of casein (Griinari et al., 1997) and in Rius et al. (2010), where casein was abomasally infused along with starch. Rulquin et al. (2004) associated the increase in MPY in response to duodenal glucose infusion with increased mammary blood flow and uptake of AA. This association between the physiological and productive responses of the mammary glands was later confirmed by Lemosquet et al. (2010) through a meta-analysis of 5 studies performing glucose infusions. Unlike the current study, in those

previous studies cows were fed at requirement levels, suggesting an effect of nutritional status on a potential interaction between AA and insulinemic effect. NASEM (2021) considers MPY to respond quadratically to MP supply, where the marginal increase in MPY diminishes as MP supply increases (Hanigan et al., 1998; Huhtanen and Hristov, 2009; Martineau et al., 2017). If insulin or glucose alters the rate of decline of those diminishing responses, a study such as ours, with limiting metabolizable AA supply, has less chance to capture any interaction between insulin and AA than studies done with cows fed at or above metabolizable AA requirements, potentially explaining the lack of synergism between insulinemic energy and mTORC1-stimulating AA in this study.

Energy Source Effects on Production and Metabolism

Treatments in this study were designed to compare insulinemic and non-insulinemic energy sources. Thus, the increase in whole-body entry rate of glucose by GLC was expected to increase pancreatic secretion and circulating concentration of insulin (Lemosquet et al., 1997), as successfully observed. Moreover, GLC increased DMI relative to ACE, contributing with additional energy and nutrients for milk production. Thus, the observed effects of GLC, relative to ACE, may be driven by nutritional or signaling effects of glucose, increased overall energy supply due to differences in DMI, signaling effects of insulin, or some combination of the 3.

As the main substrate for lactose synthesis (Scott et al., 1976) and lactose being the main osmotic regulator of milk volume (Akers, 2016), it is not surprising that GLC

Table 7. Partitioning of N losses and AA efficiency as predicted by NASEM (2021)

Item	Treatment ¹			
	GLC		ACE	
	CTL	ILM	CTL	ILM
Partition of N losses ²				
Rumen losses				
g/d	139	141	131	131
% of intake	20.6	20.7	20.1	20.2
% of total N supply	20.6	20.4	20.1	19.9
Digestive				
g/d	159	160	154	154
% of intake	23.7	23.7	23.9	23.9
% of total N supply	23.7	23.4	23.9	23.6
Catabolic				
g/d	140	153	142	152
% of intake	20.7	22.5	21.9	23.5
% of total N supply	20.7	22.2	21.9	23.2
Efficiency, ³ g/g				
Metabolizable				
Protein	0.71	0.68	0.68	0.67
Arg	0.50	0.51	0.49	0.50
His	0.83	0.84	0.81	0.82
Ile	0.63	0.58	0.61	0.57
Leu	0.73	0.59	0.71	0.58
Lys	0.65	0.66	0.63	0.64
Met	0.81	0.58	0.78	0.56
Phe	0.58	0.59	0.57	0.57
Thr	0.63	0.63	0.61	0.62
Trp	0.83	0.83	0.81	0.82
Val	0.71	0.71	0.68	0.69

¹GLC = infusions including 4.25 mol/d of glucose; ACE = infusions including 13.7 mol/d of acetate; CTL = infusions not including AA; ILM = infusions including 12, 50, and 20 g/d of Ile, Leu, and Met, respectively.

²Rumen losses = RDP balance (g)/6.25. Digestive losses = RUP + duodenal microbial CP (g) – MP not including infusions (g)/6.25. Catabolic losses = MP-N including infusions (g) – scurf true protein (TP) N (g) – fecal endogenous TP-N (g) – milk TP-N (g).

³Combined efficiency of MP or AA = (scurf TP + metabolic fecal TP + milk protein yield)/(MP or AA supply – endogenous urinary TP), according to Lapierre et al. (2020).

increased lactose and milk yield, as also observed by other studies infusing glucose postruminally (Rigout et al., 2002b; Danes et al., 2020). Unlike the current study, the lack of milk and lactose responses to glucose infusions reported by others (Lemosquet et al., 1997; Nichols et al., 2019c) are likely explained by the decrease in DMI in those glucose-infused cows. Moreover, insulin stimulates the uptake of glucose in insulin-sensitive tissues such as muscle and adipose (Bauman, 2000), potentially partitioning some glucose away from the mammary gland. However, the increases in circulating glucose and lactose yield in response to GLC suggest that mammary glucose supply was still increased by GLC in the present study.

Glucose infusion also increased milk protein content and yield, in line with other studies infusing glucose or propionate ruminally or postruminally (Rigout et al., 2003; Danes et al., 2020). Additionally, Danes et al. (2020) reported decreased concentrations of AA in plasma, particularly EAA and BCAA, supporting the idea that insulin increases mammary clearance of AA from plasma

to sustain milk protein production (Bequette et al., 2001; Cant et al., 2022), or by sequestration and catabolism by peripheral tissues (Cant et al., 2022), likely influenced by the anabolic effects of insulin.

Despite increasing lactose and protein yield, GLC decreased milk fat content, yield, and fat-corrected milk yield, further aligning with previous glucose-infusing studies (Rigout et al., 2002a; Danes et al., 2020) or insulin clamp studies (Corl et al., 2006). More specifically, GLC reduced output of $\geq C_{16}$ FA, whereas no differences were observed in $< C_{16}$ FA. Preformed FA in milk are mostly derived from dietary sources or mobilization of body fat reserves. The insulinemic nature of GLC is expected to stimulate glucose uptake, FA synthesis, and lipid accumulation in adipocytes, while also playing an inhibitory role in lipolysis (Flatt and Ball, 1964; Bauman, 2000). Therefore, by increasing insulin, GLC would have increased lipogenesis and decreased lipolysis in the adipocytes, and the availability of adipose-derived FA for milk fat synthesis, which is supported by the observed

increase in ADG and reduced circulating concentrations of NEFA relative to ACE. The decrease in NEFA and BHB along with a decrease in $>C_{16}$ FA and increase in $<C_{16}$ FA in milk are commonly reported responses to glucose or insulin infusions in lactating cows (Lemosquet et al., 1997; Rigout et al., 2002a; Corl et al., 2006; Nichols et al., 2019c). In our case, no effect of ES was observed on $<C_{16}$ FA yield, suggesting that the synthesis of longer-chain acyl-CoA from infused ACE occurred in extramammary tissues, whereas low insulin would have driven continuous release of synthesized FA, in line with higher NEFA levels in ACE compared with GLC.

We infused isoenergetic amounts of GLC and ACE to overcome the confounding effect of total energy supply, although GLC increased DMI. However, the choice of non-insulinemic energy source may also condition the relative response to GLC. In the ruminant mammary gland, acetate is the main substrate for acetyl-CoA synthesis and ATP production via the citric acid (TCA) cycle, potentially sparing glucose for lactose synthesis (Akers, 2016). Additionally, acetate is the major substrate for de novo FA synthesis, providing both carbons and reducing equivalents (Bauman et al., 1970; Urrutia and Harvatine, 2017). Abomasal ACE infusion increased acetate concentration in plasma despite the lower DMI of ACE-infused cows. Interestingly, ACE infusion did not affect the yield of $<16C$ FA synthesized de novo in the mammary glands. Hence, the lower milk fat and FCM yield response in GLC relative to ACE seems to have been driven by the insulinemic effect of glucose in adipose tissue rather than by a substrate effect of acetate. Higher concentrations of preformed FA exert an inhibitory effect on de novo FA synthesis (Neville and Picciano, 1997; Nichols et al., 2020), which could have restricted the additional acetate availability for this purpose. Others have used long-chain FA (Nichols et al., 2019a) or oils (Arieli et al., 2001) as non-glucogenic energy sources, which would not provide energy or reducing equivalents to the mammary epithelial cell, and may further inhibit de novo milk fat synthesis. As there is not a perfect control to compare the response to increased glucose supply, the characteristics of the alternative treatment used should be taken into consideration. Although the mechanisms of insulin regulation of milk component synthesis are not fully elucidated, our findings, in line with previous reports, suggest a combination of systemic and cellular effects regulating nutrient partitioning, synthetic functions, and overall nutrient efficiency for milk production, likely limited to a specific lactation stage, insulin sensitivity level, and nutritional status.

Independent Effects of AA

Beyond being essential substrates for milk protein synthesis, the group of AA infused in this study share

a specific role in translation regulation through the stimulation of mTORC1 (Arriola Apelo et al., 2014b; Pszczolkowski et al., 2020). Additionally, Met is known to play a critical role in the synthesis of the universal methyl donor S-adenosylmethionine (SAM), as well as the antioxidants glutathione and taurine (McFadden et al., 2020). Several meta-analyses have shown that Met supplementation via postruminal infusion or rumen-protected products consistently improves milk protein production (Vyas and Erdman, 2009; Patton, 2010; Zanton et al., 2014), although the marginal effects of Met decrease as the supply of metabolizable Met increases (Vyas and Erdman, 2009). Production responses to Ile and Leu supplementation have been more inconsistent. Jugular infusion of Ile plus Leu increased milk protein yield independently of the level of Met, Lys, and His (Yoder et al., 2020), suggesting complementary mechanisms of these 2 groups of AA to stimulate milk protein synthesis. Furthermore, Doelman et al. (2015) observed that removal of the 3 BCAA or just Leu from an abomasally infused EAA mix decreased milk protein yield. Conversely, Weekes et al. (2006) observed no effects of removing BCAA from a complete AA mix on milk protein yield. Similarly, jugular infusion of BCAA had negative (Curtis et al., 2018) or no effect (Appuhamy et al., 2011; Nichols et al., 2022) on milk protein yield. The lack of a positive response in the latter studies could be explained by an antagonistic effect of Val on Ile and Leu metabolism (Wang et al., 2023; Hruby Weston et al., 2024). As suggested by others (Appuhamy et al., 2011; Curtis et al., 2018; Cant et al., 2022), supplementation of Ile and Leu could also direct other circulating AA toward extramammary tissues, explaining the decrease in concentration of total EAA (except ILM), NEAA (not accounting for Cys), and total AA in circulation despite lack of change in MPY observed in ILM-infused cows. The current study was not designed to determine where and how AA not used for milk protein synthesis would be used. However, circulating concentrations of Ala and Gln, the main forms of transport of amino groups to the liver, and of UN, would increase if other AA were being deaminated in extramammary tissues (Bender, 2012). Instead, ILM treatment, despite increasing metabolizable AA supply, decreased circulating concentrations of Ala, Gln, and UN, and failed to increase MUN, urinary UN, and total urine N output. These results suggest that AA leaving the plasma pool were not deaminated for oxidation of their carbon skeletons. More likely, ILM stimulation of mTORC1 in extramammary tissues increased AA demands for protein synthesis, as reported to happen with Leu supplementation in neonatal pigs (Murgas Torrazza et al., 2010). It is important to note that ILM also increased circulating insulin, which may have further contributed to mTORC1 stimulation and protein synthe-

sis increase in muscle (Davis et al., 2002). In general, AA have been found to stimulate insulin secretion directly via depolarization of the cell membrane of pancreatic β -cells, which induces insulin exocytosis (McClenaghan et al., 1996; Newsholme et al., 2007), or indirectly via mitochondrial signals or incretin-dependent mechanisms (Gameiro et al., 2005). Additionally, studies have suggested that Leu increases insulin secretion by increasing cellular energy production in pancreatic β -cells (Fajans et al., 1967; Malaisse et al., 1980; Gao et al., 2003).

Interestingly, ILM decreased milk lactose content and yield. Even though glucose is the main precursor for lactose synthesis, some studies have shown that increased AA supply, particularly BCAA, can increase lactose yield either by sparing the use of glucose for ATP production or via signaling mechanisms that stimulate lactose synthase activity (Maxin et al., 2013; Nichols et al., 2019d; Silva et al., 2022). Curtis et al. (2018) and Rulquin and Pisulewski (2006) reported linear decrease in lactose yield in response to increasing doses of BCAA infused into the jugular vein or Leu infused into the duodenum. However, unlike in our study, neither of those studies observed changes in insulin upon AA infusion, further supporting the proposition of a direct effect of ILM on extramammary glucose uptake. Alternatively, the mammary glands of ILM-infused cows may have increased glucose channeling through glycolysis to sustain the synthesis of glycerol and the reduction of cytosolic NADP (Akers, 2016; Nichols et al., 2019d) to support the increased *de novo* FA in milk observed in ILM-infused cows. Although the effects of ILM on milk fat synthesis have not been fully elucidated, one mechanism by which AA stimulate FA synthesis is through mTORC1 regulation of SREBP-1 translocation to the nucleus for transcription regulation of key lipogenic genes (Porstmann et al., 2008; Hao et al., 2022; Li et al., 2023). In line, Met has shown to consistently increase milk fat yield, although to a lesser extent than milk protein (Patton, 2010; Zanton et al., 2014), and Daley et al. (2022) observed through a meta-analysis that absorbed Ile and Met are positive predictors of milk fat yield. However, further research on the mechanisms of *de novo* milk fat synthesis regulation is needed to design proper nutritional strategies for the optimization of milk component production.

Nitrogen Utilization and Partitioning

Dietary N supply is the major driver of N losses in lactating cows (Huhtanen and Hristov, 2009). However, energy source can affect the effect of N intake on NUE. In this study, GLC increased DM and N intake, resulting in higher predicted MP-N supply relative to ACE, but NUE was not affected. In fact, we observed a slightly higher

efficiency of MP and of all EAA for GLC compared with ACE. As discussed before, the role of glucose and insulin in stimulating milk protein synthesis likely resulted in a more efficient transfer of AA into milk. Similar increases in NUE driven by improved milk protein production in response to glucose infusions are reported by others (Danes et al., 2020; Omphalius et al., 2020).

Unlike GLC inadvertently increasing amino-N supply through DMI, the increase in metabolizable amino-N supply induced by ILM relative to CTL was by design. However, ILM failed to increase milk TP-N, while increasing N supply, leading to a decrease in MP efficiency. Several hypotheses that cannot be proven in the current study could explain the lack of response to ILM. The rather drastic drop we observed in metabolizable use efficiency of Leu and especially Met indicate a diminishing cellular response to those AA for milk protein synthesis purposes. Meanwhile, the high metabolic efficiency of His among treatments would suggest a deficiency relative to other EAA, potentially limiting milk protein synthesis and the sequestration of more abundant AA (Bequette et al., 2000; Weekes et al., 2006). Furthermore, treatments did not affect MUN, urine UN, or daily urine N output, indicating that the greater MP supply with GLC or ILM did not increase catabolism of AA. Instead, the decreases in serum UN and plasma concentrations of total AA with ILM infusions suggest that AA in excess were utilized for anabolic processes in peripheral tissues, as discussed previously. Similarly, Yoder et al. (2020) observed no effects of jugular infusions of Met, Lys, and His or Ile and Leu as separate groups on MUN or N efficiency. However, unlike in the current study, AA infusion in that study increased milk protein yield and decreased BW. Differing results were observed by Omphalius et al. (2020), where infusion of AA with the profile of casein increased catabolism of group 2 AA (Ile, Leu, Lys, and Val) by peripheral and mammary tissues. Conversely, infusion of glucose reduced mammary catabolism of group 2 AA and increased EAA efficiency, suggesting that the change of fate for AA happened within the tissues and not between them (Omphalius et al., 2020). The differences in metabolic fates for AA between studies are not captured by NASEM (2021). For example, in the current study, ILM reduced serum UN, suggesting higher post-absorptive N efficiency, whereas NASEM (2021) predicts a decrease in MP efficiency. This finding underscores the importance of refining nutrient prediction models to better represent post-absorptive N partitioning and cellular metabolism in response to individual AA supplementation. Further understanding the roles of insulin, individual AA, and energy substrates on partitioning AA toward milk protein synthesis could allow the development of nutritional strategies that potentiate the signaling roles of these molecules for improved N efficiency.

CONCLUSIONS

Contrary to our hypothesis, abomasal infusion of glucose as an insulinemic energy source did not potentiate ILM on MPY in peak-lactation cows compared with acetate, a non-insulinemic, isoenergetic energy source selected as control. Meanwhile, GLC did increase MPY, probably through a combination of effects from glucose, insulin, and the increase in DMI, leading to an increase in MP efficiency. Furthermore, the anabolic effects of insulin in GLC-infused cows are evidenced through decreased circulating NEFA, resulting in decreased milk fat yield, particularly FA >C₁₆ and lower milk fat yield and FCM relative to ACE. Unlike ES, ILM infusion did not affect DMI or MPY and decreased plasma EAA – ILM concentrations and serum UN, suggesting that AA were partitioned toward extramammary tissues for anabolic purposes to a greater extent than toward milk, thereby explaining the lack of MPY response to ILM. Fascinatingly, ILM increased de novo (<C₁₆) and tended to increase mixed (C₁₆) milk FA, suggesting a functional transductional role of Ile, Leu, and Met on FA synthesis. Neither factor affected NUE.

NOTES

This experiment was supported by the USDA-NIFA-AFRI grant 2022-67015-36197 (Washington, DC). We thank the donations of Spectrum Fusion by Perdue Agribusiness (Salisbury, MD), DL-Met by Adisseo (Alpharetta, GA), and AjiPro-I and AjiPro-L by Ajinomoto Health and Nutrition (Itasca, IL). We are grateful for the support of the staff at the University of Wisconsin–Madison Dairy Cattle Center (Madison, WI). Protocols for animal handling in this study were approved by the College of Agriculture and Life Sciences Institutional Animal Care and Use Committee at the University of Wisconsin–Madison (protocol number A006518). Authors Arriola Apelo and Danes have previously received research funds from feed companies that commercialize AA. However, no company evaluated the experimental results of the current study or had any say in their interpretation. The authors have not stated any other conflicts of interest.

Nonstandard abbreviations used: ACE = acetate infusion treatment; BCAA = branched-chain AA; CTL = control treatment, no AA infused; ES = energy source; FA = fatty acid; GLC = glucose infusion treatment; ILM = Ile, Leu, and Met infusion treatment; M- = metabolizable; M-AA = metabolizable AA; MPY = milk protein yield; mTOR = mechanistic target of rapamycin; mTORC1 = mTOR protein complex 1; NEFA = nonesterified fatty acid; NUE = nitrogen use efficiency; TG = triglyceride; TP = true protein; UN = urea N.

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